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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Major Project Phase-II (BCS786)



Stage-1 Presentation On “Multi-Modal Analysis using Spectral Biomarkers for Early Detection of Alzheimer’s Disease”

Team No: 36

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DEPARTMENT VISION

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- To promote activities that imparts high quality technical education in core domains of computer engineering and also interdisciplinary areas.
- To imbibe innovative projects and research based skills using software tools to solve real time problems for building modern India and the world.
- To create a conducive environment for lifelong learning with ethical responsibilities to produce globally competent Computer professionals and entrepreneurs.

- Project Overview (Based on work done in Phase-I)
- Methodology
- Implementation
- Testing and Validation
- Results and Discussion
- Status of Paper Publication and NPTEL Course
- Conclusion
- References

Project Overview

- Conducted an extensive literature survey on recent machine learning and deep learning approaches for early detection of Alzheimer's disease, identifying current limitations and research gaps.
- Defined the problem statement and proposed a multimodal framework that integrates EEG signals, speech data, and cognitive assessment data for improved diagnosis.
- Selected spectral biomarkers as the primary features by extracting frequency-domain characteristics from neurological signals to detect early symptoms of Alzheimer's disease.
- Identified the software stack and tools required for implementation, including Python, NumPy, Pandas, SciPy, MNE, Librosa, Scikit-learn, TensorFlow/PyTorch, and Flask/FastAPI.
- Finalized the proposed system architecture, functional requirements, non-functional requirements, and overall project workflow for implementation in the next phase.
- Planned to evaluate the model using performance metrics such as Accuracy, Precision, Recall, and F1-score, with ensemble machine learning and deep learning models to improve robustness and prediction performance.

Methodology

Step 1: Data Collection

- Collect multimodal data from publicly available datasets:
 - EEG signals
 - Speech recordings
 - Cognitive assessment scores
- These diverse data sources capture different neurological characteristics associated with Alzheimer's disease.

Step 2: Data Preprocessing

- Remove noise and artifacts from EEG signals.
- Clean and normalize speech recordings.
- Handle missing values and standardize cognitive assessment data.
- Ensure all data is in a suitable format for analysis.

Step 3: Spectral Feature Extraction

- Extract frequency-domain features (spectral biomarkers) from EEG and speech signals using signal processing techniques.
- Obtain relevant cognitive features from assessment data.
- Combine all extracted features into a unified feature set.

Methodology

Step 4: Feature Fusion

- Integrate EEG, speech, and cognitive features into a single multimodal feature vector.
- This fusion improves the model's ability to capture complementary information from different modalities.

Step 5: Model Training

- Train multiple Machine Learning and Deep Learning models such as:
 - Logistic Regression
 - Random Forest
 - XGBoost
 - LightGBM
- Develop an ensemble model to improve prediction accuracy and robustness.

Step 6: Classification

- Classify subjects into:
 - Healthy Control
 - Mild Cognitive Impairment (MCI)
 - Alzheimer's Disease (AD)

Methodology

To obtain the final diagnosis, we used an SLSQP-based weighted ensemble, assigning optimized weights to cognitive, EEG, and speech predictions.

$$P_{\text{ensemble}} = 0.34P_{\text{cognitive}} + 0.33P_{\text{EEG}} + 0.33P_{\text{speech}}$$

Adaptive Weight Optimization

- Cognitive Data → **0.34**
- EEG Data → **0.33**
- Speech Data → **0.33**

If one modality is unavailable, the system automatically redistributes the weights. For example:

- Cognitive Data → **0.55**
- Remaining Modalities → **0.45**

This adaptive weighting enables reliable predictions even when one or more inputs are missing.

Step 7: Performance Evaluation

- Evaluate the trained models using:
 - Accuracy
 - Precision
 - Recall

Implementation

➤ Frontend & User Interface

Developed using **HTML5, CSS3, JavaScript, and Responsive Styling**.

Provides an intuitive web portal for:

- User registration and patient demographic metric entry with file uploads.
- Live browser microphone speech recording using Web Audio API.
- Interactive prediction risk visualization and 1-click PDF report generation.

➤ Backend & Production Server Stack

Implemented using **Flask Web Framework** and served via **Waitress Production WSGI Server**.

Responsible for:

- RESTful request handling and input payload validation.
- Executing multi-model inference pipelines.
- User authentication and role-based access control
- Automated diagnostic PDF report generation and download.

➤ Database & Data Persistence Layer

Utilizes **SQLite3 Database** to store:

- Patient demographic and clinical assessment details.
- Historical prediction records, diagnostic scores, and generated reports.

Implementation

➤ **Modality Machine Learning Pipelines**

- **Cognitive Pipeline:** Preprocesses clinical metrics, applies Z-score feature scaling, and generates prediction.
- **EEG Signal Pipeline:** Filters 19-channel signals (0.5-45"Hz"), extracts Welch PSD features, and generates prediction.
- **Speech Pipeline:** Trims audio silence, extracts acoustic features (MFCCs, Pitch, Jitter, Shimmer), and generates prediction.

➤ **Multimodal Meta-Fusion Integration**

- Combines probability outputs from active pipelines using **Sequential Least Squares Programming** (SLSQP) weight optimization.
- Applies Adaptive Ensemble Learning to dynamically reweight modalities when optional scans are omitted, generating the final patient diagnosis.

Testing and Validation

1. Cognitive Data

Dataset: 373 patient records (OASIS Cognitive Dataset)

- Median imputation for missing values, Z-score normalization for feature scaling.
- Feature engineering performed on cognitive assessment parameters, Validated preprocessing before model training.

2. EEG Data

Dataset: 88 subjects (1,320 EEG epochs)

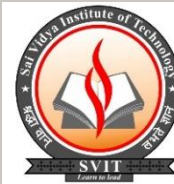
- Applied **4th-order Butterworth Bandpass Filter (0.5–45 Hz)** to remove noise.
- Segmented EEG into **2-second epochs**.
- Extracted Power Spectral Density (PSD) features using Welch's Method.
- Generated features from Delta (δ), Theta (θ), Alpha (α), Beta (β), and Gamma (γ) frequency bands.

3. Speech Data

Dataset: 591 speech recordings

- Removed background silence using audio preprocessing.
- Extracted:
 - 20 Mel Frequency Cepstral Coefficients (MFCCs)
 - YIN Pitch Contour (F_0)
 - Jitter and Shimmer features
- Selected the **Top-35 most informative features** using **SelectKBest** feature selection.

Results and Discussion



- Cognitive Monotonicity: Linear models with L2 regularization fit tabular cognitive scores (MMSE & CDR) with zero overfitting risk compared to deep neural nets.
- EEG Spectral Slowing: Gradient-boosted tree splits effectively captured non-linear frequency band shifts across 19 electrode channels.
- Acoustic Generalization: Bagging 200 decision trees over 35 selected features (MFCCs, Pitch, Jitter, Shimmer) prevented overfitting on high-dimensional speech features.
- Among the individual modalities, Logistic Regression achieved the best performance on cognitive data with an accuracy of 94.64% and an ROC-AUC of 0.9677.
- For EEG data, XGBoost produced the highest accuracy of 77.50% by effectively capturing non-linear spectral patterns.
- In the speech pipeline, Random Forest achieved 92.13% accuracy by handling a large number of acoustic features.

Status of Paper Publication and NPTEL Course



- Developed multimodal AI platform for the early prediction of **Alzheimer's Disease (AD)** and **Mild Cognitive Impairment (MCI)**.
- Integrated **Cognitive Assessments, 19-Channel EEG Signals, and Speech Biomarkers** into a unified diagnostic framework.
- Applied optimized ensemble learning to improve prediction accuracy, robustness, and reliability.
- Implemented modality-specific preprocessing and feature extraction pipelines for cognitive, EEG, and speech data.
- Developed an adaptive weighted ensemble capable of generating predictions even when one modality is unavailable.
- Combined three complementary biomarkers to overcome the limitations of single-modality diagnosis.
- Supports **early screening** and timely clinical intervention for Alzheimer's Disease.

[1] **OASIS Longitudinal Cognitive Dataset:** Washington University Alzheimer's Disease Research Center.

Source: Open Access Series of Imaging Studies. URL: <https://www.oasis-brains.org/>

[2] **OpenNeuro ds004504 19-Channel Rest EEG Dataset:** *"A 19-channel EEG dataset of 88 subjects: 36 Alzheimer's disease patients, 23 Frontotemporal dementia patients, and 29 Healthy controls"*, OpenNeuro Repository (2023).

Source: OpenNeuro. URL: <https://openneuro.org/datasets/ds004504>

[3] **DementiaBank Pitt Speech Corpus:** Clinical Speech Audio Dataset.

Source: TalkBank Repository. URL: <https://dementia.talkbank.org/>

Software Tools, Libraries & Frameworks Used

- **Programming Language & IDE:** Python 3.10.12, VS Code IDE.
- **Signal & Audio Processing:** MNE-Python (v1.12.1), Librosa (v0.10.1), SciPy, SoundFile.
- **Machine Learning & Deep Learning:** Scikit-Learn (v1.4.0), XGBoost, LightGBM, CatBoost, TensorFlow / Keras.
- **Web Framework & Database:** Flask (v3.0.0), Waitress WSGI Server, SQLite3 Database Engine.

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- [5] N. K. Kumar, V. D. A. Kumar, D. Quamar, B. S. K. Reddy, R. Yogendra, and N. Poojitha, "Enhanced Alzheimer's Disease Prediction through Advanced Imaging: A Study of Machine Learning and Deep Learning Approaches," *Proc. IEEE Int. Conf.*, pp. 1–6, 2023.
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- [7] A. Singh *et al.*, "Federated learning for privacy-preserving Alzheimer's disease prediction using cognitive and socio-demographic data," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 8, pp. 3890–3901, 2023.
- [8] S. Gayathri *et al.*, "Two-stage AI framework for EEG-based stress monitoring and Alzheimer's risk assessment," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 1420–1431, 2023.
- [9] R. Danda *et al.*, "TADEC: Explainable AI framework for preclinical Alzheimer's prediction using naturalistic driving behavior," *IEEE Access*, vol. 11, pp. 45210–45222, 2023.
- [10] C. R. Jack *et al.*, "NIA-AA Research Framework: Toward a biological definition of Alzheimer's disease," *Alzheimer's & Dementia*, vol. 14, no. 4, pp. 535–562, 2018.

Thank You